

GWANGJU AP, Korea, Republic ofWMO: **471580**Lat: **35.126N**Lon: **126.809E**Elev: **39**StdP: **14.68**Time Zone: **9.00 (E09)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	21.5	11.9	10.1	27.9	14.4	11.4	29.6	20.1	33.9	17.5	34.8	2.6	30	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	93.4	79.3	91.1	78.1	88.2	76.6	80.7	88.8	79.7	87.6	78.7	86.2	8.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	150.3	84.4	77.3	142.4	83.1	76.8	139.6	82.7	44.4	88.8	43.3	87.6	42.2	86.6	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.5	12.7	DB	11.8	95.5	3.3	2.1	9.4	97.0	7.4	98.2	5.6	99.4	3.2	100.9	(4)
(5)			WB	11.9	83.8	3.2	1.7	9.6	85.0	7.8	86.0	6.0	87.0	3.6	88.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.4	34.2	36.5	44.2	55.4	64.3	73.2	80.4	81.2	71.8	59.7	47.9	38.4
	DBStd	17.23	5.59	6.62	5.84	6.66	4.60	3.60	4.59	3.65	5.43	5.92	6.33	5.27
	HDD50	1560	491	379	191	23	0	0	0	0	0	5	111	360
	HDD65	4287	956	798	643	295	68	1	0	0	8	180	513	825
	CDD50	4258	0	1	13	185	445	696	941	966	655	306	49	1
	CDD65	1509	0	0	0	8	48	247	476	501	213	16	0	0
	CDH74	13405	0	0	0	98	458	1580	4565	5078	1476	150	0	0
	CDH80	4740	0	0	0	6	70	447	1766	2073	371	7	0	0
Wind	WSAvg	4.5	4.7	5.2	5.4	5.1	5.0	4.4	5.1	4.4	3.7	3.6	4.0	4.1
Precipitation	PrecAvg	55.1	1.3	1.8	2.6	3.6	3.9	6.5	11.6	12.4	5.8	2.1	2.2	1.5
	PrecMax	76.5	3.3	5.2	5.4	9.1	8.5	14.4	24.8	24.9	20.0	8.1	5.8	3.0
	PrecMin	32.6	0.2	0.4	0.6	0.7	1.0	1.4	2.1	3.3	0.9	0.1	0.2	0.1
	PrecStd	10.9	0.7	1.1	1.3	2.0	1.7	3.5	5.0	5.4	4.5	1.7	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	59.4	66.4	80.6	85.2	89.9	100.0	96.8	90.0	80.8	70.0	59.4
		MCWB	46.8	51.6	53.0	61.7	66.8	72.6	78.4	80.1	75.6	66.0	60.1	50.6
	2%	DB	49.9	55.1	62.6	76.9	82.1	87.6	93.7	93.6	86.4	78.4	66.6	55.4
		MCWB	45.5	47.5	50.8	61.5	64.9	71.5	79.5	79.4	73.8	64.3	56.5	47.7
	5%	DB	46.6	51.5	59.2	73.2	79.0	84.6	91.2	91.5	84.2	75.0	64.1	51.9
		MCWB	41.7	45.7	48.8	59.3	63.6	70.2	78.5	78.6	72.2	63.3	54.9	44.8
	10%	DB	43.2	48.1	55.8	69.4	76.6	82.4	88.2	89.4	81.8	71.7	60.4	48.6
		MCWB	38.9	42.4	47.0	57.5	62.8	69.8	77.7	77.5	71.3	61.2	53.6	43.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	55.0	58.4	69.1	71.1	77.7	81.9	83.0	78.5	70.1	63.2	52.5
		MCDB	50.9	57.4	61.2	74.5	77.1	83.2	90.3	88.6	83.6	74.8	66.8	58.0
	2%	WB	46.1	49.8	53.0	65.1	68.6	75.7	80.5	81.2	76.5	67.3	59.4	49.5
		MCDB	48.5	52.7	60.1	71.6	76.7	80.2	89.3	88.5	83.4	73.3	64.5	53.0
	5%	WB	42.2	45.9	50.5	61.4	66.3	74.2	79.5	80.1	75.0	65.4	56.7	46.2
		MCDB	45.3	50.4	56.3	70.0	74.6	79.4	87.9	87.9	81.3	72.0	61.1	50.7
	10%	WB	39.7	42.7	48.2	58.6	64.7	72.4	78.6	79.2	73.2	63.1	54.2	43.6
		MCDB	43.1	47.4	55.0	67.2	72.7	78.8	86.4	86.7	79.8	70.0	59.7	48.3
Mean Daily Temperature Range	5% DB	MDBR	14.9	16.9	20.2	23.4	21.4	16.3	13.2	15.0	17.9	22.4	20.8	18.2
		MCDBR	18.3	21.4	26.4	28.8	27.4	22.7	17.6	18.1	19.1	25.4	26.5	23.4
		MCWBR	14.2	15.1	16.3	15.0	13.0	9.8	7.0	7.2	8.7	13.4	17.0	16.5
	5% WB	MCDBR	15.8	19.2	21.9	22.9	20.5	15.1	15.0	15.2	15.0	21.0	21.2	22.0
		MCWBR	13.6	14.6	15.5	14.0	11.6	8.4	7.0	7.4	8.3	12.4	15.5	16.8
Clear-Sky Solar Irradiance	taub		0.456	0.496	0.569	0.586	0.603	0.666	0.621	0.569	0.506	0.479	0.465	0.455
	taud		1.917	1.855	1.713	1.723	1.735	1.679	1.830	1.956	2.064	2.002	1.970	1.929
	Ebn at Noon		221	229	224	229	228	213	222	231	240	232	217	211
	Edh at Noon		47	56	71	73	73	77	66	57	49	48	45	44
All-Sky Solar Radiation	RadAvg		770	1022	1383	1601	1747	1558	1438	1479	1322	1173	827	681
	RadStd		60	91	123	128	188	144	197	204	131	102	88	47